

SYED MUHAMMAD ALI GILANI

Electrical Engineer | PEC Registered Engineer

Satellite Town, Rawalpindi, Pakistan | +92-336-0777949

syedmuhammadaligillani111@gmail.com | linkedin.com/in/syedmuhammad-ali-gilani

PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in Electronics with a strong academic foundation and practical industrial exposure to electrical maintenance, hydropower generation, substation operations, and transformer testing. Knowledgeable in preventive maintenance, equipment troubleshooting, electrical safety (LOTO), and power distribution systems. Proficient in MATLAB, Simulink, and technical documentation. Seeking an entry-level Maintenance Engineer position to contribute to equipment reliability and efficiency.

EDUCATION

Bachelor of Electrical Engineering

Federal Urdu University of Arts, Science and Technology (FUUAST), Islamabad

CGPA: 3.22/4.00

PROFESSIONAL EXPERIENCE

Power Systems Trainee – 32 MW Hydropower Project

- Gained practical exposure to hydropower generation processes and plant operations.
- Developed understanding of turbine, synchronous generator, excitation and governor systems.
- Learned Emergency Shutdown (ESD) procedures and Lockout/Tagout (LOTO) practices

Transformer Testing & Diagnostics Trainee – Hammad Engineering & Transformer Co.

- Developed understanding of transformer oil testing, dielectric strength, moisture analysis and DGA.
- Reviewed transformer condition assessment reports and preventive maintenance practices.
- Assisted in diagnostic procedures used to improve equipment reliability.

CORE COMPETENCIES

- Preventive Maintenance
- Corrective Maintenance
- Equipment Troubleshooting
- Electrical Systems Maintenance
- Power Distribution Systems
- Transformer Testing & Diagnostics
- Electrical Safety (LOTO)
- Technical Documentation

TECHNICAL SKILLS

- Engineering Software: MATLAB, Simulink, Proteus, AutoCAD, Arduino IDE, Microsoft Office
 - Engineering Knowledge: Electrical Maintenance, Preventive & Corrective Maintenance, Power Distribution Systems, Electrical Machines, Transformers, High Voltage Engineering, Equipment Troubleshooting
-

FINAL YEAR PROJECT

GNSS/INS-Based Vehicle Navigation Prototype

- Designed and developed an Arduino Uno-based vehicle navigation prototype integrating GPS and IMU sensors.
 - Integrated hardware components and implemented embedded control logic using Arduino IDE.
 - Performed system calibration, functional testing, troubleshooting, and technical documentation.
 - Verified system performance through hardware integration and prototype validation
-

PROFESSIONAL DEVELOPMENT & LEADERSHIP

- Organizer – YOUTH ROBOTECH
 - Participant – AI INNOFEST (Bahria University)
 - Participant – National Youth Conference (PICSS)
 - Workshops: GNSS/INS Navigation, Line Follower Robot, Home Automation, RC Plane
-

LANGUAGES

English | Urdu